Arista Access Node KR-ACC-56 QoS Congestion Event Remediation SOP

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5427

Description

The failure domain of QoS/congestion events (microbursts, interface tail drops, QoS queue starvation) on the Arista Access node KR-ACC-56 is a critical concern that necessitates meticulous troubleshooting and resolution. This SOP outlines the diagnostic and resolution procedures to address the Alarm Code ALARM_5427, which indicates a warning-level condition. The underlying protocol mechanics responsible for this failure domain are rooted in the complex interplay between Quality of Service (QoS) policies, traffic policing, and queuing mechanisms. Specifically, the Arista Access node's role in shaping and policing traffic, combined with its reliance on hardware-assisted policing and queuing, can lead to congestion events when traffic bursts exceed the configured policers or when QoS queues become starved due to excessive traffic load. Microbursts, which are sudden spikes in traffic, can further exacerbate this condition by overwhelming the policers and causing interface tail drops. To rectify this situation, it is essential to employ a structured approach that involves thorough diagnostic procedures to identify the root cause, followed by targeted resolution steps to mitigate the congestion event.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and the node KR-ACC-56 is powered on.

- Check the redundant link status to ensure that it is operational and not contributing to the congestion event.

- Gather relevant logs and configuration files from the node to aid in troubleshooting.

Potential Root Causes

Cause	Probability
Inadequate policer configuration	High
Insufficient QoS queue depth	Medium
Microbursts causing interface tail drops	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface kr-acc-56 | include policer, Expected Output: policer 1000p 1000p 1000p 1000p, Description: This command is used to verify the policer configuration on the interface kr-acc-56. The expected output indicates that the policer is configured to police traffic at 1000 pps.
```

```
Command: show qos kr-acc-56 | include queue-depth, Expected Output: queue-depth 1000, Description: This command is used to verify the QoS queue depth on the node kr-acc-56. The expected output indicates that the queue depth is set to 1000 packets.

Command: show interface kr-acc-56 | include tx-queue-drops, Expected Output: tx-queue-drops 0, Description: This command is used to verify the number of transmit queue drops on the interface kr-acc-56. The expected output indicates that there are no transmit queue drops.
```

Resolution Steps

Step 1: Step: Adjust policer configuration, Command: config interface kr-acc-56 policer 2000p 2000p 2000p 2000p, Description: This step involves adjusting the policer configuration to allow for more traffic to be policed. The command updates the policer configuration to police traffic at 2000 pps.

Step 2: Step: Increase QoS queue depth, Command: config qos kr-acc-56 queue-depth 2000, Description: This step involves increasing the QoS queue depth to prevent queue starvation. The command updates the queue depth to 2000 packets.

Step 3: Step: Verify policer and QoS queue configuration, Command: show interface kr-acc-56 | include policer && show qos kr-acc-56 | include queue-depth, Description: This step involves verifying that the policer and QoS queue configuration have been updated correctly.

Post-Check Verification

```
Command: show interface kr-acc-56 | include policer && show qos kr-acc-56 | include queue-depth, Expected Output: policer 2000p 2000p 2000p 2000p && queue-depth 2000, Description: This command is used to verify that the policer configuration and QoS queue depth have been updated correctly.

Command: show interface kr-acc-56 | include tx-queue-drops, Expected Output: tx-queue-drops 0, Description: This command is used to verify that there are no transmit queue drops on the interface kr-acc-56.
```

Roll-back Procedures

- Step: Revert policer configuration, Command: config interface kr-acc-56 policer 1000p 1000p 1000p 1000p, Description: This step involves reverting the policer configuration to its original value. The command updates the policer configuration to police traffic at 1000 pps.

- Step: Revert QoS queue depth, Command: config qos kr-acc-56 queue-depth 1000, Description: This step involves reverting the QoS queue depth to its original value. The command updates the queue depth to 1000 packets.